

Using SQL in Python to Analyze Sales Data

This script connects to a SQLite database, runs SQL queries to extract sales summary information (total quantity sold and total revenue), prints the results, and visualizes the output using a basic bar chart.

Python Script:

```
import sqlite3
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Connect to the SQLite database
```

```
conn = sqlite3.connect("sales_data.db")
```

```
# Run SQL query to get total quantity sold and total revenue
```

```
query = """
```

```
SELECT
```

```
    SUM(quantity) AS total_quantity_sold,
```

```
    SUM(quantity * price) AS total_revenue
```

```
FROM sales
```

```
"""
```

```
df = pd.read_sql_query(query, conn)
```

```
conn.close()
```

```
# Extract values
```

```
total_quantity_sold = df['total_quantity_sold'][0]
```

```
total_revenue = df['total_revenue'][0]
```

```
# Print the results
```

```
print("=== Basic Sales Summary ===")
```

```
print(f"Total Quantity Sold: {total_quantity_sold}")
```

```
print(f"Total Revenue: ${total_revenue:.2f}")
```

```
# Plot the summary as a bar chart
```

```
df.T.plot(kind='bar', legend=False, color=['green'])
```

```
plt.title("Total Sales Summary")
```

```
plt.ylabel("Amount")
```

```
plt.xticks(rotation=0)
```

```
plt.tight_layout()
```

```
# Save the chart
```

```
plt.savefig("total_sales_summary_chart.png")
```

```
plt.show()
```

To run this script:

1. Ensure 'sales_data.db' is in the same directory.
2. Run with: `python sales_summary_with_chart.py`